EE604 COURSE PROJECT / 2026

Cricket Shot Recognition



Gradient features and a frozen CNN

A controlled four-class image study

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https://github.com/hrushikeshroop/cricket-shot-recognition-ee604

Problem, labels and constraints

Input: one selected cricket image. Output: one of four dataset labels.

Drive

Leg glance / flick

Pull

Sweep



CPU inference, offline after setup.

No batter detection, contact localization or technique grading.

Unknown shots can still receive confident scores.

Course foundation and prior work

Use established feature extraction, then test one modest extension

Foundation	Source	Our use
Gradient histograms	Lecture 10; Dalal / Triggs (2005)	Explicit HOG baseline
Frozen pretrained features	Lecture 15, pp. 11-14	MobileNetV2 + linear classifier
Cricket recognition	KUCricShot (2023); Kang (2025)	Prior task studies; different data
Two-view difference	Course implementation	Compare with simple concatenation

No claim of a new backbone or an unprecedented combination. Novelty remains limited.

Images and fixed visual-group split

4,724 original files; 4,723 exact representatives in 363 visual groups

Class	Train	Validation	Test
Drive	734	244	245
Glance / flick	672	224	224
Pull	756	252	252
Sweep	672	224	224

221 training groups / 72 validation groups / 70 test groups

No shared group or decoded hash. Match and player identities remain unknown.

Evaluation choices and their limits

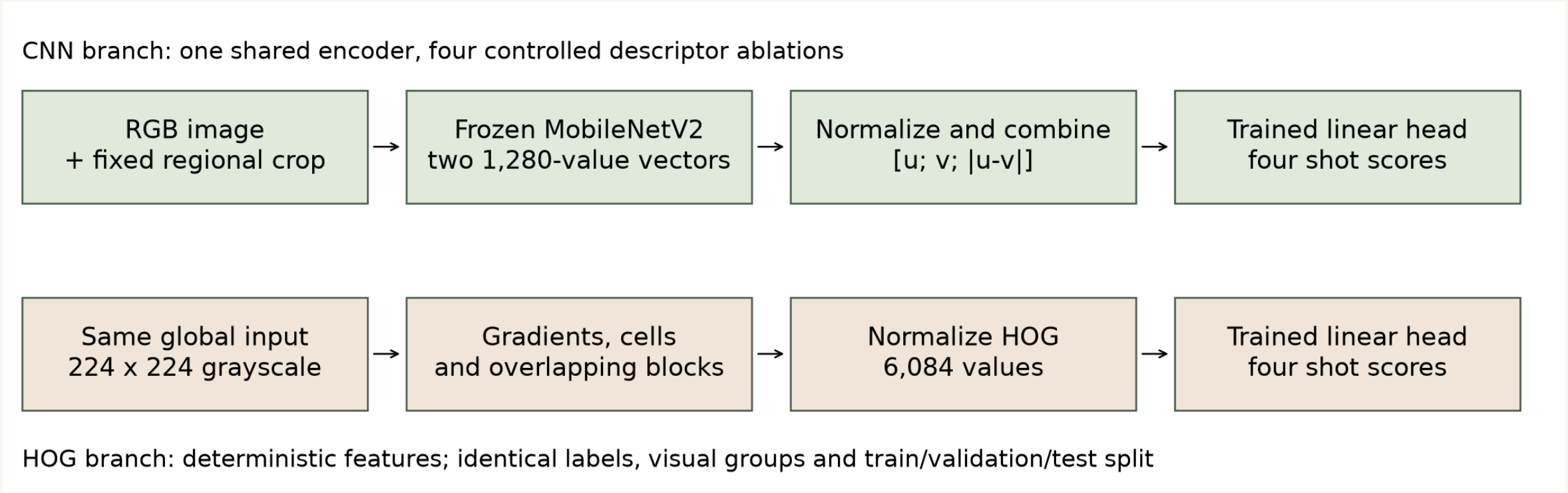
All five methods share images, labels, visual groups and split identities

Choice	Data used	What remains fixed
Fit classifier coefficients	Training only	Frozen CNN; deterministic HOG
Choose C: 0.1, 1, 10, 100	Validation macro F1	Smallest C wins ties
Temperature and review	Original validation reuse	Keep original settings
Final comparison	Same 945 test identities	No new split or seed search

Exploratory: grouping changed after preliminary evaluation; HOG was added after old results were known.

Implemented pipeline

Only the small cricket classifiers learn from the training images

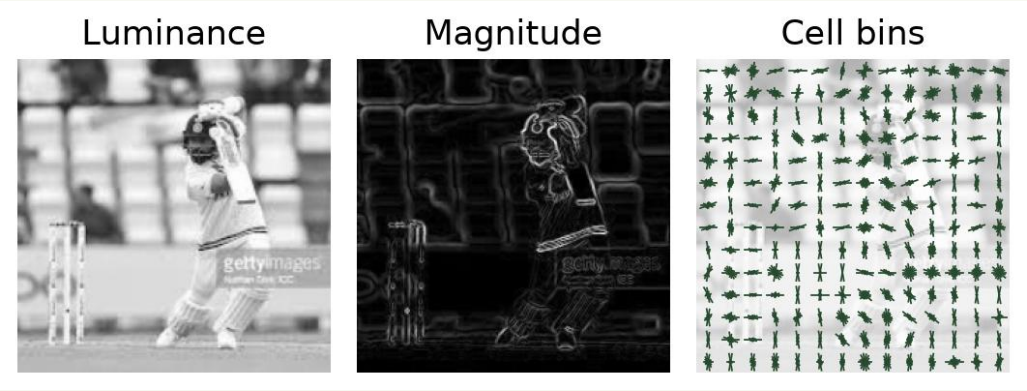


The fixed crop is a regional view. It does not detect or segment the batter.

HOG: gradients, cells and blocks

Lecture 10: accumulate gradient magnitude in orientation bins

Step	Fixed setting
Input	224 x 224 luminance
Orientation votes	9 unsigned bins; linear bin weights
Cells	16 x 16 pixels; 14 x 14 cells
Blocks	2 x 2 cells; stride 1 cell
Normalize / concatenate	L2 blocks, then 6,084-value L2 vector



Votes use magnitude, not frequency. No learned feature scaling; no test-based HOG tuning.

CNN views and the contrast feature

Frozen ImageNet MobileNetV2 gives two 1,280-value pooled embeddings

Representation	Normalized classifier input	Dimension
Global	u	1,280
Fixed regional crop	v	1,280
Concatenation	[u; v]	2,560
Contrast	[u; v; abs(u-v)]	3,840

Absolute difference is nonlinear; signed difference would be redundant for a linear head on concatenation. This motivates a test, not a success claim.

Recognition results

Same corrected split; every test image contributes, including review cases

Feature	C	Val. F1	Test acc.	Test F1
HOG	0.1	49.4%	35.3%	34.5%
CNN global	10	77.1%	63.0%	62.9%
CNN crop	10	74.8%	62.4%	61.7%
Concatenation	10	79.6%	65.2%	64.6%
Contrast	10	77.8%	65.5%	64.9%

HOG is a controlled baseline. Pretrained features help here; contrast adds only three correct images.

Uncertainty comes from 70 visual groups

Whole-group bootstrap, 2,000 paired draws, fixed model and split

Feature	Accuracy	Conditional 95% interval
HOG	35.3%	25.9 to 48.1%
CNN global	63.0%	49.5 to 74.2%
CNN crop	62.4%	45.3 to 76.7%
Concatenation	65.2%	48.9 to 77.9%
Contrast	65.5%	49.9 to 77.6%

Contrast minus concatenation: +0.32 points; interval -1.46 to +2.38. No clear advantage.

Class errors

Contrast confusion counts; rows are dataset labels

True / predicted	Drive	Glance/flick	Pull	Sweep
Drive	161	27	36	21
Glance / flick	80	102	15	27
Pull	36	13	197	6
Sweep	1	15	49	159

Macro F1 averages the four class F1 scores equally. Review does not remove mistakes from this table.

Scores, calibration and secondary review

Temperature scaling worsened the recorded test calibration

Contrast scores	NLL	ECE
Unscaled	0.859	0.043
Scaled	0.898	0.094

557 / 945 accepted

433 accepted predictions correct

58.9% coverage / 77.7% accuracy

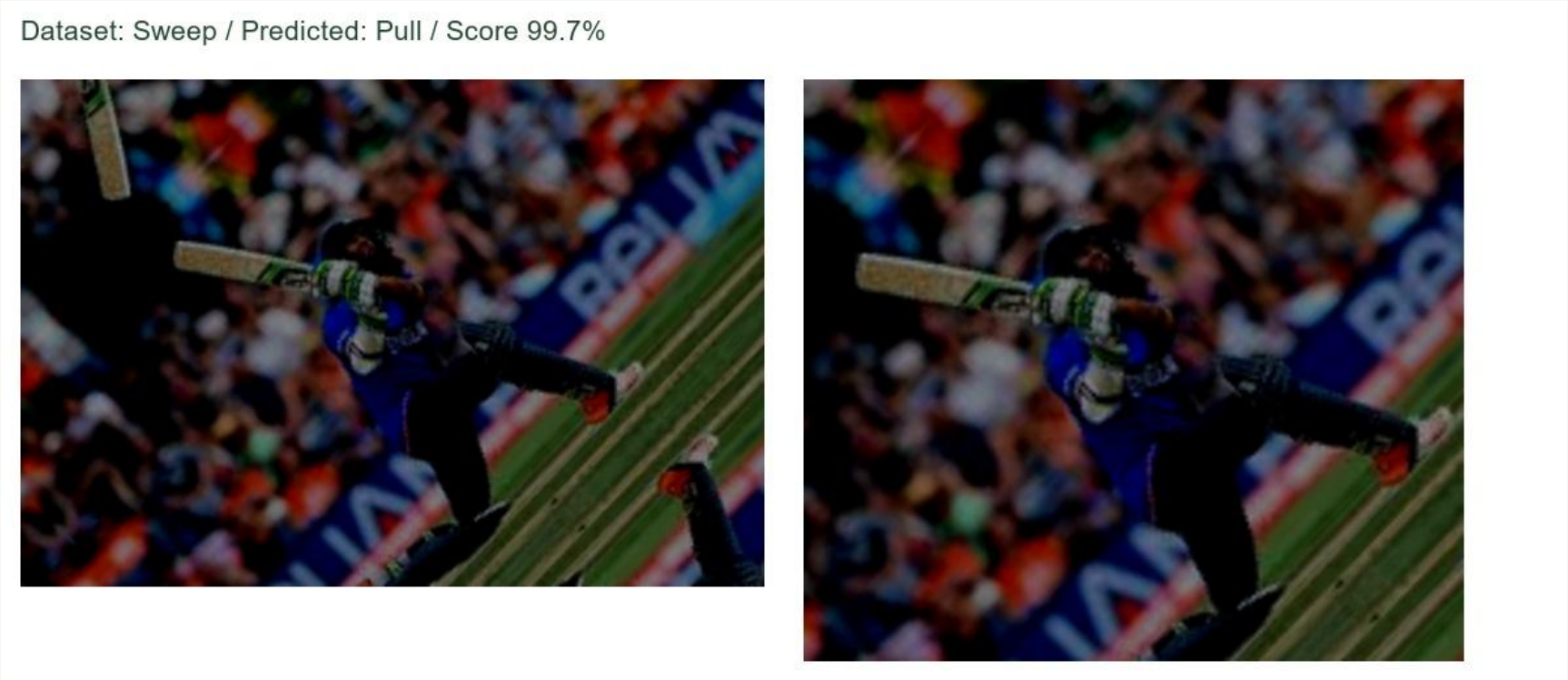
Keep T = 0.802 and threshold = 0.70 unchanged.

The 90% validation target did not transfer to test.

Softmax is a relative class score. Agreement is not a guarantee or an unknown-shot detector.

A confident error stays in the demo

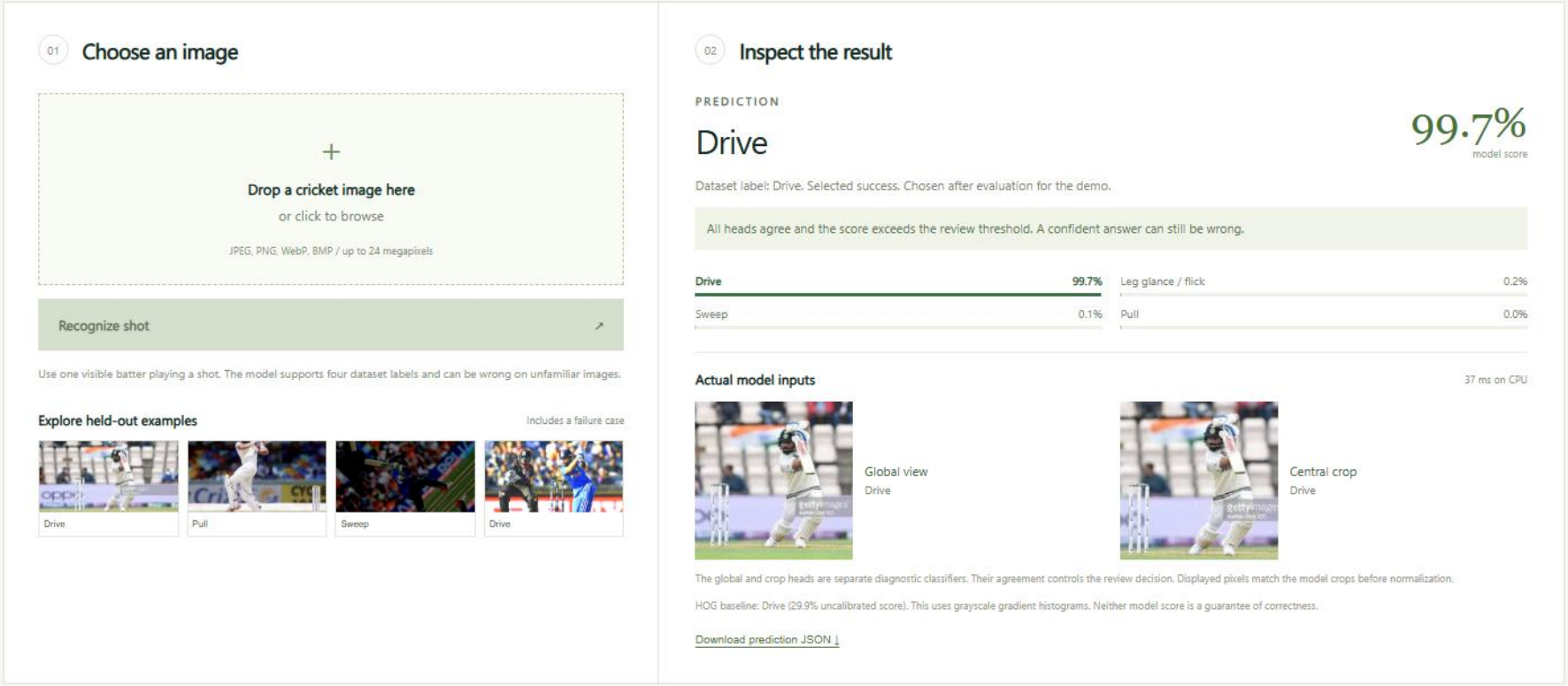
Selected after evaluation, alongside successful and disagreement examples



Inspect the upstream label, predicted label and actual input views. Confidence can be wrong.

Local image demonstration

Upload an image, compare CNN and HOG, inspect the review reason, export JSON



What we can defend

An implemented comparison with useful negative findings

Course foundation

HOG and frozen features, each followed by a trained multiclass head.

Limited extension

Contrast gains three test images; uncertainty spans no gain.

The audit corrects optimism, and calibration fails to transfer.

Next useful step: independently labeled data with known match and player identities.

Backup: video pilot and duplicate audit

Separate eight-class data and reused pilot holdout; not the main recognition claim

Video model	Accuracy	Macro F1
Full frame	18.1%	17.7%
Fixed crop	30.2%	21.3%
Majority baseline	30.2%	5.8%

562 clips / 25 source groups / 116 test clips from five groups

Image audit: hashes, transforms, ORB/RANSAC links; all connected copies stay together.